

Rabeya Khan

✉ rabeyakhan592@gmail.com ☎ 01633911374 📍 Dhaka, Bangladesh 🏠 Rabeya Khan

👤 PROFILE

Motivated Electrical and Electronic Engineering graduate with strong research experience in renewable energy systems, machine learning, and computational modeling. Skilled in literature review, data analysis, and manuscript preparation. Experienced in funded research projects, IoT, and embedded systems, with proven ability in teamwork, scientific writing, and research collaboration through first-author publications and mentoring activities. The long-term goal is to build a strong career in advanced research and technical teaching.

🎓 EDUCATION

B.Sc. Engineering in Electrical and Electronic Engineering <i>Jamalpur Science & Technology University (JSTU)</i> CGPA: 3.81/4.00 (Class position: 2nd)	01/2020 – 09/2025 Jamalpur, Bangladesh
Higher Secondary Certificate (H.S.C) <i>Manikganj Govt. Mohila College</i> Science Background GPA: 5.00/5.00	07/2017 – 04/2020 Manikganj, Bangladesh
Secondary School Certificate (S.S.C) <i>Prattasha Adarsha High School</i> Science Background GPA: 4.68/5.00	01/2015 – 04/2017 Dhaka, Bangladesh

🧰 PROFESSIONAL EXPERIENCE

Research Assistant <i>Robo Tech Valley</i> - Conducting IoT and Machine Learning-based Research - Preparing Journal Manuscripts and Conference Papers	05/2026 – Present Dhaka, Bangladesh
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★ CAPSTONE PROJECTS

Bridging the Lab-to-Field Gap: Attention-Consistent Vision Transformers with Faithfulness-Aware Explanation for Robust Rice Disease Diagnosis <i>Research Assistant</i> Developed a Vision Transformer-based rice disease detection framework with attention-consistency learning for robust field deployment under domain shift.	07/2026 – 07/2026 Dhaka, Bangladesh
A Multimodal Framework for Livestock Disease Screening Using Multi-Repository Visual Datasets and Symptom-Based Intelligence <i>Research Assistant</i> Developed a multimodal livestock disease screening system combining YOLOv8 object detection, dataset harmonization, and symptom-based clinical assessment.	06/2026 – 06/2026 Dhaka, Bangladesh
AquaSense IoT: An Cloud-Integrated Framework for Real-Time Water Quality Monitoring and Control Using pH, DO, TDS, Temperature, and Water Level Sensors <i>Research Assistant</i> Developed a two-tier IoT system for real-time water quality monitoring, cloud communication, and automated environmental control using Arduino, ESP32, and Firebase.	05/2026 – Present Dhaka, Bangladesh
A Biogas Power Plant Integrated with IoT-based Monitoring System and Machine Learning-Based analysis (Funded by RIC GSTU) <i>Investigator</i> Developing an IoT-enabled monitoring system for biogas and electricity generation with real-time data acquisition.	09/2024 – 03/2025 Jamalpur, Bangladesh
Design, Set up and Analysis of Off-Grid Solar System (Funded by Renewable Energy Research Center, JSTU) <i>Co-mentor</i> Co-mentored the design and performance analysis of a 5 kW AC/DC off-grid solar power system.	13/05/2024 – 26/05/2024 JSTU, Bangladesh

★ RESEARCH EXPERIENCE

Undergraduate Research Assistant

06/2022 – 09/2025

Supervisor: Dr. Md. Rasidul Islam

- Computational modeling and efficiency enhancement of lead-free perovskite solar cells.
- Machine learning-based performance analysis of perovskite solar cells.
- First-principles simulations of perovskite materials using Quantum ESPRESSO.
- Writing, formatting and editing manuscripts for journal submission.

📖 PUBLICATIONS

● **Rabeya Khan**, Nadira Farjana, Mst. Jahida Akter Jim, Jehan Y. Al-Humaidi, Md Rasidul Islam, Md Masud Rana, "Numerical simulation and performance enhancement of $\text{CsBi}_3\text{I}_{10}$ -based heterojunction solar cell", *Solar Energy*, 295, 113539, 2025. (Elsevier) <https://doi.org/10.1016/j.solener.2025.113539>  22/04/2025

● **Rabeya Khan**, I. K. Gurusul Ghosh Apurba, Rahat Ul Nasib, Md Rasidul Islam-PhD, "Device Modeling and Performance Enhancement of Ecologically Viable CsGeI_3 -Based Perovskite Solar Cells", *Nano Select*, 2025. (Wiley) <https://doi.org/10.1002/nano.70026>  27/05/2025

● I.K. Gusrul Ghosh Apurba, Md. Rasidul Islam, **Rabeya Khan**, Md Rahat Ul Nasib, Md. Ferdous Rahman, Farhad Ilahi Bakhsh, Abu Zahid, Sohail Ahmad, "The Density Function Theory Analysis of Mg_3AsX_3 Perovskite Materials: Strain Effect Study", *Materials Science in Semiconductor Processing*, 183, 108741, 2024. (Elsevier) <https://doi.org/10.1016/j.mssp.2024.108741>  15/11/2024

★ CONFERENCE PRESENTATION

5th International Conference on "Physics for Sustainable Development & Technology (ICPSDT 2023)"

09/2023

Presented the work titled 'Design and Performance Analysis of Lead-Free Perovskite Solar Cell Using SCAPS-1D Device Simulation'.

Chittagong University
of Engineering and
Technology

★ EXTRACURRICULAR ACTIVITIES

President - Jamalpur Science & Technology University Debating Society

05/2025 – 11/2025

- Led the JSTU Debating Society by organizing and presiding over university-level debates, workshops, and training programs; represented the university in inter-university competitions, fostering a strong culture of public speaking, critical thinking, and collaborative argumentation while enhancing the society's reputation.

Jamalpur, Bangladesh

🧩 WORKSHOPS AND ACADEMIC ACTIVITIES

Teaching Assistant, Communication Engineering Laboratory

09/2024 – 08/2025

Assisted in conducting undergraduate laboratory sessions by delivering theory lectures on basic communication techniques (amplitude, frequency DSB-SC, and SSB-SC modulation, demodulation) and supervising students throughout laboratory experiments, including equipment setup, operation, and result analysis.

Dept. of EEE, JSTU

Instructor, Fundamental Robotics Workshop

03/2023

Conducted hands-on workshops on Arduino fundamentals and Arduino programming, guiding students through the implementation of simple projects, including sensor control and data acquisition.

Robotics Club, JSTU

Instructor, Parliamentary Debate Workshops

10/2024 – 09/2025

Designed and conducted multiple workshops on parliamentary debate, focusing on argument structure, critical thinking, and public speaking.

JSTU Debating Society

🏆 AWARDS AND HONORS

Merit Scholarship Award 2023

2025

Jamalpur Science & Technology University (JSTU)

Merit Scholarship Award 2022 <i>Jamalpur Science & Technology University (JSTU)</i>	2024
Merit Scholarship Award 2021 <i>Jamalpur Science & Technology University (JSTU)</i>	2023
10th Prize-Grand Finale of Farm Mechanism Idea Competition 2025 <i>Bangladesh Agricultural Research Institute, Gazipur</i>	2025
First Position – Innovation Showcasing Programme <i>E-Governance and Innovation Committee, JSTU</i>	2024
Third Position – Poetry Recitation Competition, JSTU on National Children's Day	2024
Second Position – Extempore Speech Competition, JSTU on National Children's Day	2024
Champion – Project Showcasing, National Conference on Robotics and IoT, Dept. of CSE, JSTU	2023
Champion-Project Showcasing, JSTU EEE Day	2022
Champion - Idea Contest, JSTU EEE Day	2022
First Position - Public Speaking Competition, JSTU on National Children's Day	2022
Champion and Best Speaker - DIUDC Bitorko Mohaparbon 1426 (Bengali) <i>Daffodil International University Debating Club (DIUDC)</i>	2019



SKILLS

Programming Languages

Assembly Language, C, Python, MATLAB, Arduino

Simulation & Analysis Tools

SCAPS 1D, OriginPro, Quantum ESPRESSO, Simulink, AutoCAD, LOGO! Soft Comfort

Hardware Platforms

Raspberry Pi, Arduino

Soft Skills

Public speaking, Presentation, Leadership, Teamwork, Communication



LANGUAGES

Bengali

Native



English

IELTS Academic: 7.0 overall (no band below 6.0)



REFERENCES

Dr. Md. Rasidul Islam, *Assistant Professor*, Dept. of EEE, Jamalpur Science & Technology University
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Md. Shizer Rahman, *Chairman*, Dept. of EEE, Jamalpur Science & Technology University
shizer@jstu.ac.bd, 01793781079